

Switching-Aware Multilayer Graph Learning for Energy-Efficient User Association and Cell Switch-ON/Off in Dense Wireless Networks

Abstract—Dense 5G and beyond-5G radio access networks require user association policies that reduce infrastructure energy consumption while preserving quality-of-service (QoS), load feasibility, and operational stability. Static BS switch-off can save energy, but frequent on/off transitions introduce additional cost and instability that are often ignored by snapshot-wise association policies. This paper formulates the problem as a multilayer network-control task and proposes a switching-aware multilayer graph-learning framework for joint user association and BS switch-off. Each wireless snapshot is modeled as a UE-level multilayer graph with separate relation layers for association competition, interference similarity, load-pressure similarity, and temporal traffic similarity. Fixed-fusion and attention-weighted multilayer graph convolutional networks (ML-GCNs) learn UE-to-BS association preferences. A score-guided active-cell pruning stage and a stability-aware pruning stage then convert learned preferences into compact active-BS sets under QoS/load repair. The framework is analyzed in terms of permutation equivariance, aggregation-induced information loss, feasibility-preserving repair, and switching-adjusted energy. In a ten-seed simulation study, the proposed stability-aware ML-GCN improves both static and switching-adjusted energy relative to greedy sleep control while maintaining full service feasibility. Stability-aware ML-GCN search achieves 79.0% static energy saving and 76.1% switching-adjusted saving relative to the all-on network, compared with 76.9% and 68.3% for greedy sleep. The gain is accompanied by fewer BS state changes, reducing active-cell churn from 2.50 to 0.85 mean transitions per snapshot. Runtime measurements further show that stable learned policies have lower online latency than greedy sleep and much lower latency than finite active-cell search. These results support multilayer graph learning combined with stability-aware pruning as a practical framework for feasible, energy-efficient, and switching-aware wireless control.

Index Terms—Graph neural networks, multilayer networks, network science, network control, user association, energy-efficient wireless networks, base-station switch-off.

I. INTRODUCTION

DENSE radio access networks are expected to support rapidly increasing traffic demand while reducing the energy footprint of wireless infrastructure. This tension is especially visible in dense 5G and beyond-5G deployments, where operators deploy macro, micro, pico, and indoor small-cell layers to improve coverage and hotspot capacity. Such densification improves service during busy periods, but it also creates many lightly loaded or temporarily redundant cells when traffic is spatially uneven or time varying. Prior green HetNet studies therefore treat base-station (BS) ON/OFF control as a radio-resource-management problem: redundant small cells may be switched to a low-power state, and their users may be served by nearby active BSs, provided rate and load constraints remain satisfied [1]–[3]. This paper follows

that infrastructure-energy viewpoint. The BSs are not assumed to be battery-limited devices; instead, the radio access network (RAN) is an energy-cost- and carbon-sensitive infrastructure whose active radio units, power amplifiers, cooling systems, and baseband resources should be used only when they provide service value [4], [5].

The relevant operational question is therefore not simply which BS gives each UE the strongest signal. It is a joint control question: which BSs should remain active, which UEs should be associated with them, and how should the active-cell set evolve over time without violating service constraints. This decision can be implemented by a network-side control entity, such as a centralized SON function, cloud/edge RAN controller, or O-RAN RIC application, using UE measurements, channel estimates, traffic demand, BS load, and previous BS states. O-RAN energy-saving work identifies RF-channel reconfiguration, advanced sleep modes, and cell/carrier shut-down as network energy-saving features coordinated through RIC/management interfaces [5]. In this setting, a BS sleep command is feasible only if users that would otherwise be served by that BS can be reassigned to active cells while maintaining radio feasibility, rate demand, and load limits.

This coupling makes energy-aware user association more difficult than association-only control. Since a BS consumes non-negligible static power even when lightly loaded, energy-efficient network operation requires the controller to reason about the global consequence of activating or deactivating cells. It also requires temporal stability. A policy that independently minimizes energy at each snapshot can increase BS state changes, and these transitions may incur signaling, hardware, and service-continuity costs. Energy-efficient control should therefore be evaluated using both static power reduction and the stability of the active-cell sequence.

User association is therefore a central control problem in dense wireless networks. Classical association rules based on received signal strength, signal-to-interference-plus-noise ratio (SINR), or load balancing are computationally efficient, but they are local by construction. They typically assign each UE using the quality or load of a small number of candidate BSs and do not evaluate the global energy consequence of activating or deactivating BSs. Optimization-based formulations can model the joint association and switch-off problem more directly, but the resulting mixed discrete-continuous decision space is difficult to solve repeatedly at network time scales. Learning-based methods can amortize computation after training; however, the quality of a learned policy depends strongly on the graph representation through which the wireless system is presented to the model.

Graph neural networks (GNNs) are suitable for wireless resource management because wireless networks are relational systems. A UE is not an isolated sample: its best association depends on neighboring UEs, competing BSs, interference coupling, traffic demand, and load created by other association decisions. Recent wireless GNN studies show that graph-based models can provide scalable approximations for resource-allocation tasks and can exploit permutation structure in wireless graphs [6], [7]. Nevertheless, many graph-learning formulations reduce the system to a single relation graph. Although aggregation simplifies model construction, it can obscure a structural distinction: a dense wireless snapshot contains multiple kinds of relations whose semantics are not interchangeable. Two UEs may compete for the same serving BS, exhibit similar traffic demand, experience correlated interference conditions, or follow similar temporal traffic patterns. These relations may overlap, but they are not equivalent and should not necessarily be represented by a single averaged graph.

These considerations motivate a multilayer network formulation. In multilayer network models, a system is represented by nodes, edges, and layers that encode different interaction types, time slices, modalities, or couplings [8]–[10]. This distinction has operational implications. Prior work on multilayer networks has shown that aggregation can remove structural information, distort centrality rankings, and obscure layer-specific organization [11], [12]. For energy-aware wireless control, this issue is consequential. If association competition, interference, load pressure, and temporal traffic similarity are collapsed into one graph, a learning model may receive an averaged relation that no longer identifies which physical or operational mechanism caused a dependency between UEs. A policy trained on such an aggregate graph may therefore reproduce associations without explicitly representing the relations that determine whether particular BSs can be switched off without violating quality-of-service (QoS) and load constraints.

From a network-science perspective, the problem is a controlled multilayer network in which layer structure shapes feasible system behavior. The nodes are UEs, the layers encode distinct wireless interaction mechanisms, and the control action changes the active infrastructure projection through BS switch-on/off decisions. The analysis therefore examines how preserving or collapsing relation layers affects downstream network control, rather than restricting the evaluation to association-label prediction.

In this paper, we study energy-efficient user association and BS switch-off as a multilayer graph decision problem. For each wireless snapshot, we construct a UE-level multilayer graph

$$\mathcal{G} = (\mathcal{V}, \mathbf{X}, \{\mathbf{A}^{(\ell)}\}_{\ell \in \mathcal{L}}), \quad (1)$$

where $\mathcal{V}$ is the set of UEs, $\mathbf{X}$ contains UE radio and traffic features, and each layer $\mathbf{A}^{(\ell)}$ represents a distinct wireless relation. The layer set $\mathcal{L}$ is defined to preserve four operational dependencies that are relevant to energy-aware association:

- the *association layer*, which connects UEs that compete for the same strongest or candidate BS;

- the *interference layer*, which links UEs with similar channel profiles and interference exposure;
- the *load layer*, which captures demand similarity and potential load-pressure coupling among UEs; and
- the *temporal layer*, which represents spatial-traffic similarity under time-varying demand.

Fig. 2 illustrates how these relation layers are constructed, processed by layer-specific graph filters, and fused before association and active-cell decisions. The final decision is evaluated by network energy, QoS satisfaction, BS activity, and load feasibility rather than by assignment accuracy alone.

The central research question is:

Can preserving multilayer wireless structure improve energy-aware user association and BS switch-off decisions, compared with conventional association heuristics, sleep-control baselines, and single-layer graph-learning models, while maintaining QoS, load feasibility, and active-cell stability?

The proposed framework is intended for execution at the network controller, not at individual UEs. The controller observes each traffic snapshot, constructs the multilayer UE graph, predicts association preferences, selects a compact active-BS set, and then applies a feasibility repair before the decision is sent to the RAN as association and sleep-mode control. The framework has two linked components. First, a multilayer GNN predicts UE-to-BS association preferences using layer-specific graph propagation. Second, an active-cell selector uses the learned association distribution and BS-level radio/load summaries to identify a compact active BS set. A QoS- and load-aware repair procedure then converts the learned preferences into a feasible association whenever possible, activating additional BSs only when the predicted active set cannot support the traffic realization. This design separates representation learning from constraint enforcement: the GNN learns multilayer structure, while the repair stage protects wireless feasibility. We further introduce score-guided pruning and stability-aware pruning policies. The score-guided policy removes low-score BSs only when energy and feasibility improve. The stability-aware policy uses the previous snapshot's active set as inertia and accepts a BS state change only when the switching-adjusted objective improves or feasibility requires the transition.

The paper makes the following contributions.

- 1) We formulate joint user association and BS switch-off as a multilayer network decision problem, with separate layers for association competition, interference coupling, load pressure, and temporal traffic similarity. The formulation preserves layer-specific wireless dependencies and admits theoretical analysis of permutation equivariance, aggregation-induced information loss, layer redundancy, and feasibility-preserving decision repair. The formulation connects BS sleep control with multilayer network reducibility and layer-aware control.
- 2) We develop fixed-fusion and attention-weighted multilayer GNN models for UE association preferences and combine them with score-guided and stability-aware active-cell pruning. The resulting policies explic-

itly trade off static BS energy, switching cost, QoS satisfaction, and load feasibility.

- 3) We define a comparative metric suite for constrained wireless control, including static energy saving, switching-adjusted energy saving, energy gap relative to a finite-search reference, served ratio, active BS count, BS state changes, maximum load, and mean active load. Layer ablation, attention entropy, relation-redundancy, and runtime measurements are included to assess modeling and operational tradeoffs.
- 4) We conduct a simulation-based comparative study against conventional RSRP, SINR, and load-aware association heuristics, explicit sleep-control heuristics, greedy BS sleep control, finite active-cell search, feature-only learning, aggregated GCN learning, fixed multilayer GCN learning, and attention-weighted multilayer GCN learning. The results distinguish static-energy performance from switching-aware operation.

Table I summarizes the main structural and control components of the proposed framework. It highlights the connection between the multilayer wireless representation, the learned decision mechanism, and the evaluation evidence used throughout the paper.

The remainder of this paper is organized as follows. Section II reviews wireless user association, graph learning for resource management, and multilayer graph modeling. Section III presents the system model and energy-aware association objective. Section IV develops the proposed multilayer graph-learning framework and its mathematical properties. Section V describes the simulation design, baselines, and metrics. Section VI reports and discusses the comparative results. Section VII concludes the paper.

II. RELATED WORK

Energy-aware user association has been widely studied as a mechanism for improving spectral efficiency, load distribution, and infrastructure energy consumption in dense wireless networks. Conventional association rules based on received signal strength, SINR, or load provide simple and scalable decisions, but they are usually myopic with respect to BS switch-off because the activation cost of each BS is not directly optimized. Optimization-based methods can incorporate energy and service constraints explicitly, but the joint association and activation problem is combinatorial and must be solved under changing channel and traffic conditions.

BS ON/OFF switching has a long history in green cellular networking. Dynamic switching methods exploit the fact that cellular traffic varies across time and space, so BSs dimensioned for peak demand may be underutilized during off-peak periods [2]. In HetNets, redundant small cells can be placed in sleep mode and users can be offloaded to active macro, small-cell, or femtocell resources when QoS remains feasible [1]. Surveys on BS ON/OFF switching in 5G further emphasize that dense, heterogeneous deployments create energy-saving opportunities, but also introduce coverage, interference, handover, and coordination challenges [3]. The present paper uses this literature as the physical motivation for the active-cell

decision: BS switch-off is not a UE-side battery-saving action, but an infrastructure energy-management action applied by the RAN to reduce static and load-dependent network power.

BS switch-off methods are often evaluated snapshot by snapshot, but practical cell activation also has a temporal dimension. Repeatedly changing the active BS set can introduce signaling overhead, hardware transition cost, and service instability. The present work therefore treats BS state changes as an explicit metric and evaluates switching-adjusted energy in addition to static network power. This distinction is central to the comparison with greedy sleep-control baselines.

GNNs provide a learning-based alternative for resource-management problems in which the wireless topology varies across network realizations. Random-edge GNNs have been used to approximate wireless resource allocation over changing interference graphs [6], while scalable GNN architectures have been analyzed for radio resource management with permutation properties and size generalization [7]. Graph attention has also been applied to wireless user association [13]. These studies support the use of graph learning for wireless control, but they typically represent the network through a single graph or a single dominant relation.

Multilayer network models provide a formal way to represent systems with multiple edge semantics, time slices, or interaction channels [8]–[10]. The importance of preserving layer structure is supported by work on structural reducibility and multilayer node ranking, where aggregation can remove relevant information or change structural rankings [11], [12]. These results are relevant to dense wireless networks because association competition, interference, load, and temporal traffic similarity describe different dependencies among UEs.

Recent multilayer-network studies further support this layer-aware viewpoint. Multilayer community detection has been studied through semi-supervised factorization and contrastive network augmentation [14], [15]; key-node identification in multilayer networks has been improved by fusing relationship strength, node features, influence range, and layer importance [16], [17]; and augmented-network dissimilarity has been used to compare network structure across categories [18]. These studies treat layer structure, node importance, and network dissimilarity as primary objects of analysis. The present work applies the same layer-aware modeling principle to an engineered wireless control problem, where the controller must infer which UEs jointly determine active-cell feasibility and switching behavior.

Several recent GNN studies have extended graph learning to multilayer or multi-view settings. The mGNN framework generalizes GNN operations to multilayer networks [19]. Dynamic-multilayer GCNs have been used to combine intralayer and interlayer convolution for industrial monitoring [20], while attention-based dynamic multilayer GNNs have been developed for temporal multilayer prediction tasks [21]. MVMA-GCN further uses multi-view attention and relationship dissimilarity to learn from multiple node-relation graphs [22]. These works demonstrate the value of layer-aware graph learning, but they do not address energy-aware wireless association with active-cell control. The present work focuses on this setting by coupling a multilayer UE representation with BS

TABLE I
STRUCTURAL AND CONTROL COMPONENTS OF THE PROPOSED FRAMEWORK

Modeling aspect	Wireless-network instantiation	Decision mechanism	Evaluation evidence
Multilayer structure	UE-level multilayer graph with association, interference, load, and temporal relation layers	Layer-specific graph filters and fixed/attention fusion preserve heterogeneous edge semantics	Aggregation non-injectivity theorem, layer dissimilarity, and layer ablation
Feasible behavior	UE association, active-BS set, load distribution, and served ratio under traffic snapshots	Learned association preferences plus QoS/load repair convert graph predictions into feasible network states	Served ratio, active BS count, maximum load, and energy gap metrics
Switching-aware control	BS switch-on/off action changes the active infrastructure projection over time	Score-guided and stability-aware active-cell policies trade static energy against switching cost	Switching-adjusted energy saving and BS state-change measurements
Scalable implementation	Controller must be efficient enough for repeated RAN operation	Feed-forward multilayer GNN approximates finite active-cell search before feasibility repair	Runtime and online-complexity comparison against greedy sleep and reference search

switch-off, QoS/load repair, and energy-oriented evaluation.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network Snapshot

Consider a dense downlink control region managed by a network-side controller. The region contains a set of controllable serving cells $\mathcal{B} = \{1, \dots, B\}$ and a set of UEs $\mathcal{U} = \{1, \dots, N\}$ observed at traffic snapshot t . In a HetNet, the controllable set may correspond to small cells under an overlay macro layer; in a dense single-tier deployment, it may correspond to a cluster of neighboring BSs. In both cases, the abstraction is the same: the controller can choose an active subset of BSs and can reassociate UEs among active candidates, subject to radio, rate, and load feasibility. The position, traffic demand, and radio measurements of UE i are represented by a feature vector $\mathbf{x}_i \in \mathbb{R}^F$, and the snapshot feature matrix is $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times F}$. Let $d_i(t)$ denote the traffic demand of UE i in Mbps. The large-scale and small-scale channel gain from BS b to UE i is denoted by $g_{i,b}(t)$, and the BS transmit power is P_{tx} .

The user association and BS switch-off decision is described by two binary variables. Let

$$z_{i,b} = \begin{cases} 1, & \text{if UE } i \text{ is associated with BS } b, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

and let $a_b \in \{0, 1\}$ indicate whether BS b is active. Each UE is associated with exactly one BS:

$$\sum_{b \in \mathcal{B}} z_{i,b} = 1, \quad \forall i \in \mathcal{U}, \quad (3)$$

and a UE can be assigned only to an active BS:

$$z_{i,b} \leq a_b, \quad \forall i \in \mathcal{U}, b \in \mathcal{B}. \quad (4)$$

B. SINR, Rate, and Load

For a given active-BS vector $\mathbf{a} = [a_1, \dots, a_B]^T$, the downlink SINR that UE i would obtain from BS b is

$$\gamma_{i,b}(\mathbf{a}) = \frac{a_b P_{\text{tx}} g_{i,b}}{\sigma^2 + \sum_{c \in \mathcal{B}, c \neq b} a_c P_{\text{tx}} g_{i,c}}, \quad (5)$$

where σ^2 is the receiver noise power. The corresponding achievable rate is modeled as

$$R_{i,b}(\mathbf{a}) = W \log_2 (1 + \gamma_{i,b}(\mathbf{a})), \quad (6)$$

where W is the system bandwidth. The RSRP feasibility condition is

$$P_{\text{tx}} g_{i,b} \geq \Gamma_{\text{rsrp}}, \quad (7)$$

and the minimum-rate condition is

$$R_{i,b}(\mathbf{a}) \geq R_{\min}. \quad (8)$$

The load contribution of UE i to BS b is the fraction of radio resources needed to support its demand:

$$\ell_{i,b}(\mathbf{a}) = \frac{d_i}{R_{i,b}(\mathbf{a})}. \quad (9)$$

The total load of BS b under association matrix $\mathbf{Z}$ is

$$\rho_b(\mathbf{Z}, \mathbf{a}) = \sum_{i \in \mathcal{U}} z_{i,b} \ell_{i,b}(\mathbf{a}), \quad (10)$$

and the load constraint is

$$\rho_b(\mathbf{Z}, \mathbf{a}) \leq \rho_{\max}, \quad \forall b \in \mathcal{B}. \quad (11)$$

C. Energy Model

The network energy model separates the static cost of keeping a BS active, the load-dependent cost of serving traffic, and the sleep-mode power of inactive BSs. This follows the BS power-modeling view used in green cellular-network studies, where sleep mode reduces but does not necessarily eliminate site power because some circuitry, cooling, monitoring, or wake-up functionality may remain active [1], [2]. The power consumed by BS b is

$$P_b(\mathbf{Z}, \mathbf{a}) = a_b (P_{\text{fix}} + P_{\text{dyn}} \rho_b) + (1 - a_b) P_{\text{sleep}}, \quad (12)$$

where P_{fix} is the fixed active power, P_{dyn} is the dynamic load-dependent power coefficient, and P_{sleep} is the sleep-mode power. The total network power is

$$P_{\text{net}}(\mathbf{Z}, \mathbf{a}) = \sum_{b \in \mathcal{B}} P_b(\mathbf{Z}, \mathbf{a}). \quad (13)$$

The all-on reference power is obtained by setting $a_b = 1$ for all $b \in \mathcal{B}$. This reference is used to report energy saving:

$$\text{ES} = \frac{P_{\text{all}} - P_{\text{net}}}{P_{\text{all}}}. \quad (14)$$

For temporally ordered snapshots, static energy alone is incomplete because active-cell transitions can introduce operational cost. Let $\mathbf{a}(t)$ be the active-BS vector at snapshot t . The number of BS state changes is

$$S(t) = \|\mathbf{a}(t) - \mathbf{a}(t-1)\|_0, \quad (15)$$

where $S(1) = 0$ for the first evaluated snapshot. With a per-transition switching cost κ , the switching-adjusted power is

$$P_{\text{sw}}(t) = P_{\text{net}}(t) + \kappa S(t). \quad (16)$$

The corresponding switching-adjusted energy saving is

$$\text{ES}_{\text{sw}} = \frac{P_{\text{all}} - P_{\text{sw}}}{P_{\text{all}}}. \quad (17)$$

When $\kappa = 0$, (17) reduces to the static energy-saving metric in (14).

D. Optimization Problem

The ideal energy-aware association problem is to minimize network power while satisfying association, activation, QoS, and load constraints:

$$\begin{aligned} \min_{\mathbf{Z}, \mathbf{a}} \quad & P_{\text{net}}(\mathbf{Z}, \mathbf{a}) \\ \text{s.t.} \quad & \sum_{b \in \mathcal{B}} z_{i,b} = 1, \quad \forall i \in \mathcal{U}, \\ & z_{i,b} \leq a_b, \quad \forall i \in \mathcal{U}, b \in \mathcal{B}, \\ & z_{i,b} P_{\text{tx}} g_{i,b} \geq z_{i,b} \Gamma_{\text{rsrp}}, \quad \forall i \in \mathcal{U}, b \in \mathcal{B}, \\ & z_{i,b} R_{i,b}(\mathbf{a}) \geq z_{i,b} R_{\min}, \quad \forall i \in \mathcal{U}, b \in \mathcal{B}, \\ & \rho_b(\mathbf{Z}, \mathbf{a}) \leq \rho_{\max}, \quad \forall b \in \mathcal{B}, \\ & z_{i,b} \in \{0, 1\}, a_b \in \{0, 1\}. \end{aligned} \quad (18)$$

This problem is combinatorial because the association matrix and active-BS vector are discrete and coupled through SINR, rate, load, and power. The goal of the proposed learning framework is therefore not to solve (18) exactly at every snapshot, but to learn a scalable policy that approximates low-energy feasible decisions from the multilayer wireless graph representation. In the switching-aware setting, the same feasible association variables are evaluated using $P_{\text{sw}}(t)$ in (16), which makes the active-cell sequence $\{\mathbf{a}(t)\}$ part of the operational objective.

IV. PROPOSED MULTILAYER GRAPH LEARNING FRAMEWORK

A. Overview

The proposed framework converts each wireless snapshot into a multilayer UE graph, learns UE-to-BS association preferences using layer-specific graph filters, predicts a compact active-BS set, and applies a QoS/load repair step before evaluation. Fig. 1 gives the computational pipeline, while Fig. 2 gives the detailed multilayer model architecture. The

key modeling choice is that different wireless relations are not merged before graph propagation. Instead, each relation layer is processed by a dedicated graph filter and fused only after layer-wise representation learning.

B. Multilayer Wireless Graph Construction

For each snapshot, let $\mathbf{G} \in \mathbb{R}^{N \times B}$ denote the channel-gain matrix with entries $g_{i,b}$, and let $\mathbf{r}_i$ be the position of UE i . The proposed model builds a multilayer graph

$$\mathcal{M} = (\mathcal{U}, \mathbf{X}, \{\mathbf{S}^{(\ell)}\}_{\ell \in \mathcal{L}}), \quad \mathcal{L} = \{a, i, l, t\}, \quad (19)$$

where a, i, l , and t denote association, interference, load, and temporal layers, respectively. Each layer is constructed as a nonnegative adjacency matrix $\mathbf{A}^{(\ell)}$ and converted into a row-stochastic support matrix

$$\mathbf{S}^{(\ell)} = (\mathbf{D}^{(\ell)})^{-1} (\mathbf{A}^{(\ell)} + \mathbf{I}), \quad (20)$$

where $\mathbf{D}^{(\ell)}$ is the diagonal row-sum matrix of $\mathbf{A}^{(\ell)} + \mathbf{I}$. The identity term preserves each UE's own features during propagation.

The association layer connects UEs that compete for the same strongest BS. Let

$$b_i^* = \arg \max_{b \in \mathcal{B}} g_{i,b}. \quad (21)$$

Then

$$A_{j,k}^{(a)} = \mathbf{1}\{b_j^* = b_k^*\}, \quad j \neq k. \quad (22)$$

The interference layer compares normalized channel-gain profiles. Define $\mathbf{q}_j = \mathbf{g}_j / \|\mathbf{g}_j\|_1$, where $\mathbf{g}_j$ is the j th row of $\mathbf{G}$. The interference similarity is

$$A_{j,k}^{(i)} = \mathbf{q}_j^T \mathbf{q}_k, \quad j \neq k. \quad (23)$$

The load layer links UEs with similar traffic demand within the same association region:

$$A_{j,k}^{(l)} = A_{j,k}^{(a)} \exp\left(-\eta_l \frac{|d_j - d_k|}{d_{\max}}\right), \quad (24)$$

where $\eta_l > 0$ controls demand-similarity sharpness and $d_{\max}$ is the maximum demand scale. Finally, the temporal layer captures spatial-demand similarity modulated by the traffic period:

$$A_{j,k}^{(t)} = h(t) \exp\left(-\eta_r \frac{\|\mathbf{r}_j - \mathbf{r}_k\|_2}{R}\right) \exp\left(-\eta_d \frac{|d_j - d_k|}{d_{\max}}\right), \quad (25)$$

where R is the cell-area radius and $h(t)$ is a bounded periodic traffic factor. Sparse top- K thresholding is applied to the similarity layers to avoid dense message passing and to emphasize dominant local dependencies.

C. Layer-Aware Association Learning

Given $\mathbf{X}$ and the supports $\{\mathbf{S}^{(\ell)}\}$, the fixed-fusion multilayer GCN computes one hidden representation per layer:

$$\mathbf{H}^{(\ell)} = \phi\left(\mathbf{S}^{(\ell)} \mathbf{X} \mathbf{W}^{(\ell)}\right), \quad \ell \in \mathcal{L}, \quad (26)$$

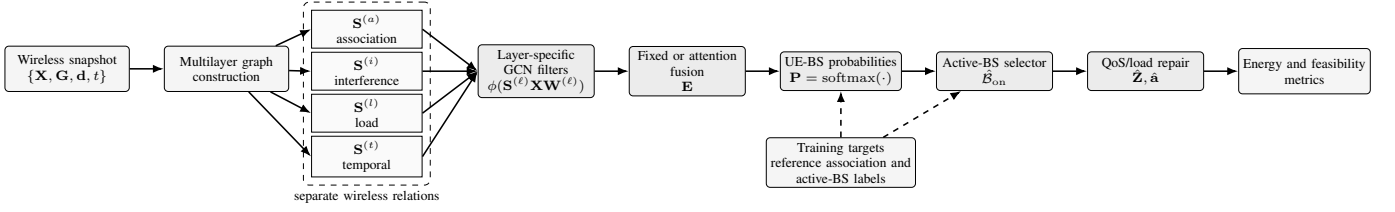


Fig. 1. Computational pipeline of the proposed framework. Wireless snapshots are converted into separate relation layers, processed by layer-specific GCN filters, fused into UE embeddings, and passed to association, active-BS selection, repair, and evaluation modules.

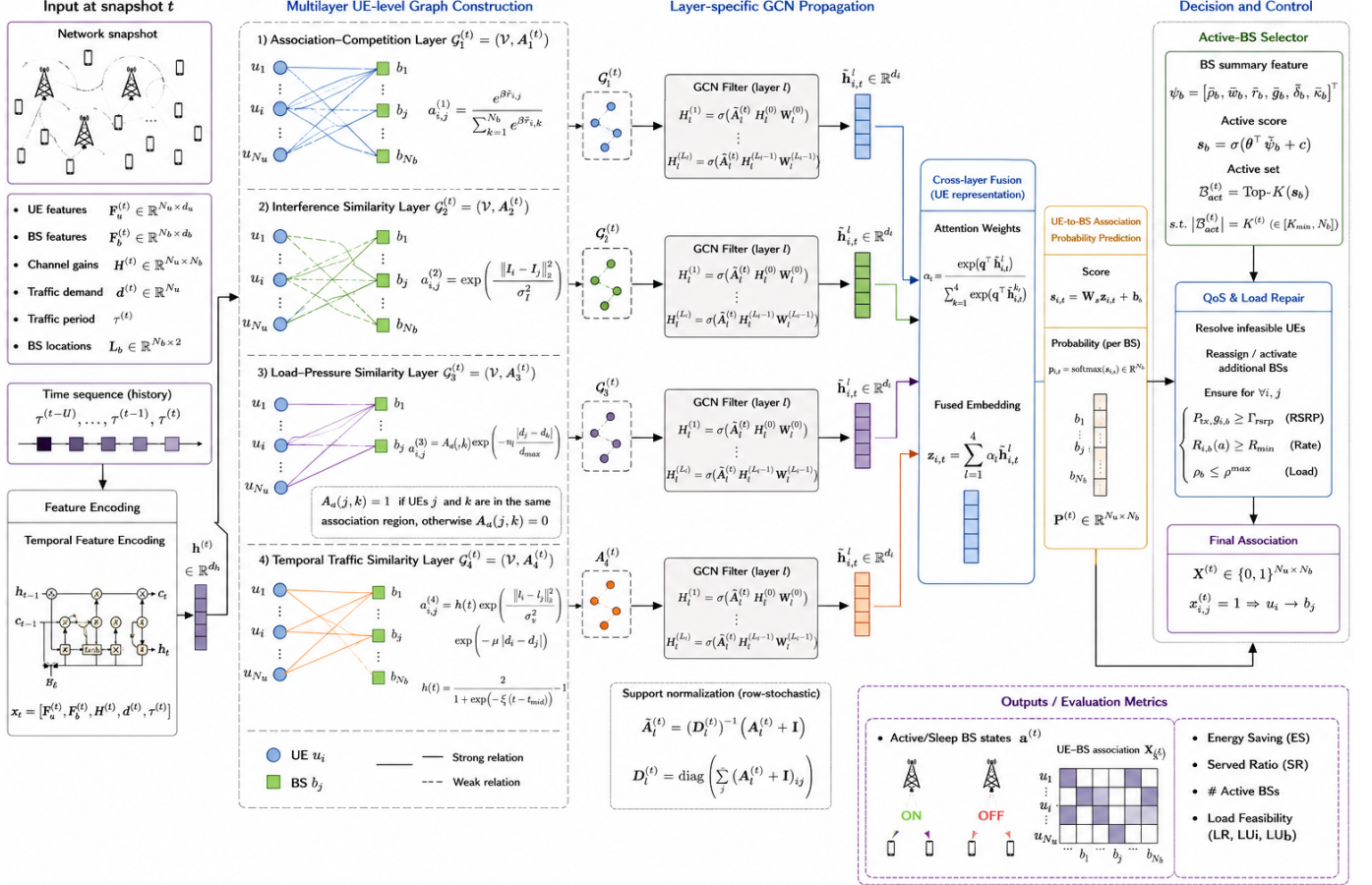


Fig. 2. Proposed multilayer graph-learning framework for energy-aware user association and BS switch-off. At each wireless snapshot, UE features, BS information, channel gains, traffic demand, and temporal context are encoded and used to construct four UE-level wireless relation layers: association-competition, interference-similarity, load-pressure similarity, and temporal-traffic similarity. Each relation layer is processed by a dedicated GCN filter to produce layer-wise UE embeddings, which are then combined through fixed or attention-based cross-layer fusion. The fused UE representation is used to predict UE-to-BS association probabilities, select a compact active-BS set, and apply QoS/load repair and report any residual infeasibility through the served-ratio metric. The final outputs include the UE-BS association matrix, active/sleep BS states, energy saving, served ratio, active-BS count, and load-feasibility metrics.

where $\phi(\cdot)$ is a pointwise nonlinearity and $\mathbf{W}^{(\ell)}$ is a layer-specific trainable matrix. The layer-wise embeddings are fused as

$$\mathbf{E} = [\mathbf{H}^{(a)} \parallel \mathbf{H}^{(i)} \parallel \mathbf{H}^{(l)} \parallel \mathbf{H}^{(t)}], \quad (27)$$

where $\parallel$ denotes feature concatenation. The UE-to-BS association probability matrix is

$$\mathbf{P} = \text{softmax}(\mathbf{E} \mathbf{W}_o + \mathbf{b}_o), \quad (28)$$

where $P_{i,b}$ is the learned probability that UE i should be associated with BS b .

The attention-weighted model assigns a trainable nonnegative importance weight to each layer. Let η_ℓ be the attention logit of layer ℓ . The normalized layer weight is

$$\alpha_\ell = \frac{\exp(\eta_\ell)}{\sum_{r \in \mathcal{L}} \exp(\eta_r)}. \quad (29)$$

The fused embedding becomes

$$\mathbf{E}_{\text{att}} = \left[|\mathcal{L}| \alpha_a \mathbf{H}^{(a)} \parallel |\mathcal{L}| \alpha_i \mathbf{H}^{(i)} \parallel |\mathcal{L}| \alpha_l \mathbf{H}^{(l)} \parallel |\mathcal{L}| \alpha_t \mathbf{H}^{(t)} \right]. \quad (30)$$

The factor $|\mathcal{L}|$ keeps the scale comparable to fixed fusion when the attention distribution is uniform.

The association predictor is trained by supervised imitation of low-energy reference decisions:

$$\mathcal{L}_{\text{ua}} = -\frac{1}{N} \sum_{i \in \mathcal{U}} \sum_{b \in \mathcal{B}} y_{i,b} \log P_{i,b}, \quad (31)$$

where $y_{i,b}$ is the reference association label. This loss trains the GNN to learn association preferences; feasibility and switch-off behavior are handled by the active-cell and repair stages below.

D. Active-BS Selection and QoS/Load Repair

Association probabilities alone do not determine which BSs should remain active, because a model may assign every UE while still keeping unnecessary BSs on. Therefore, the proposed framework uses a second-stage active-BS selector. For each BS b , a summary vector ψ_b is formed from probability statistics, demand-weighted probability mass, approximate load, channel quality, and distance statistics. The active score is

$$s_b = \sigma \left(\theta^T \bar{\psi}_b + c \right), \quad (32)$$

where $\bar{\psi}_b$ is the standardized BS summary vector and $\sigma(\cdot)$ is the logistic sigmoid. The selected active set is the set of top-scoring BSs,

$$\hat{\mathcal{B}}_{\text{on}} = \text{TopK}_{b \in \mathcal{B}}(s_b). \quad (33)$$

The value of K is estimated from the average active-BS count of the reference solutions in the training set.

The initial UE assignment chooses, for each UE, the most likely BS among $\hat{\mathcal{B}}_{\text{on}}$. A repair routine then checks the RSRP, rate, and load constraints in (7)–(11). If the selected active set cannot support the snapshot, additional BSs are activated according to their aggregate gain-demand score over stressed UEs. Algorithm 1 gives the complete inference procedure.

E. Mathematical Properties

Let $\mathfrak{M} = (\mathbf{X}, \{\mathbf{S}^{(\ell)}\}_{\ell \in \mathcal{L}})$ denote the multilayer input and define the layer-aware encoder F_{Θ} by (26)–(30). For a permutation matrix Π , define the action

$$T_{\Pi}(\mathfrak{M}) = \left(\Pi \mathbf{X}, \{ \Pi \mathbf{S}^{(\ell)} \Pi^T \}_{\ell \in \mathcal{L}} \right). \quad (34)$$

Define also the feasibility predicate

$$\Phi(\mathbf{Z}, \mathbf{a}) = 1 \iff (3), (4), (7), (8), (11) \text{ hold for } (\mathbf{Z}, \mathbf{a}). \quad (35)$$

Theorem 1 (Equivariance of the multilayer encoder). *For any permutation matrix Π ,*

$$F_{\Theta}(T_{\Pi}(\mathfrak{M})) = \Pi F_{\Theta}(\mathfrak{M}). \quad (36)$$

Algorithm 1 Multilayer GNN Association With Active-BS Repair

Require: $(\mathbf{X}, \mathbf{G}, \mathbf{d}, t)$, parameters $\Theta = \{\mathbf{W}^{(\ell)}, \mathbf{W}_o, \mathbf{b}_o, \eta_{\ell}\}$, selector (θ, c) , K , fusion mode $q \in \{\text{fixed}, \text{adaptive}\}$

Ensure: $\hat{\mathbf{Z}}, \hat{\mathbf{a}}$

```

1:  $\mathcal{L} \leftarrow \{a, i, l, t\}$ 
2:  $\mathbf{S}^{(\ell)} \leftarrow \text{LayerSupport}_{\ell}(\mathbf{X}, \mathbf{G}, \mathbf{d}, t)$ ,  $\ell \in \mathcal{L}$ 
3: for  $\ell \in \mathcal{L}$  do
4:    $\mathbf{H}^{(\ell)} \leftarrow \phi(\mathbf{S}^{(\ell)} \mathbf{X} \mathbf{W}^{(\ell)})$ 
5: end for
6: if  $q = \text{fixed}$  then
7:    $\mathbf{E} \leftarrow \mathbf{H}^{(a)} \parallel \mathbf{H}^{(i)} \parallel \mathbf{H}^{(l)} \parallel \mathbf{H}^{(t)}$ 
8: else
9:    $\alpha_{\ell} \leftarrow \exp(\eta_{\ell}) / \sum_{r \in \mathcal{L}} \exp(\eta_r)$ ,  $\ell \in \mathcal{L}$ 
10:   $\mathbf{E} \leftarrow \parallel_{\ell \in \mathcal{L}} (|\mathcal{L}| \alpha_{\ell} \mathbf{H}^{(\ell)})$ 
11: end if
12:  $\mathbf{P} \leftarrow \text{softmax}(\mathbf{E} \mathbf{W}_o + \mathbf{b}_o)$ 
13: for  $b \in \mathcal{B}$  do
14:    $\psi_b \leftarrow \text{BSFeatures}(b, \mathbf{P}, \mathbf{G}, \mathbf{d})$ 
15:    $s_b \leftarrow \sigma(\theta^T \psi_b + c)$ 
16: end for
17:  $\mathcal{A} \leftarrow \text{TopK}_{b \in \mathcal{B}}(s_b, K)$ 
18: repeat
19:    $z_{i,b} \leftarrow 0$ ,  $\forall i, b$ 
20:   for  $i \in \mathcal{U}$  do
21:      $\mathcal{C}_i \leftarrow \{b \in \mathcal{A} : R_{i,b}(\mathcal{A}) \geq R_{\min}, P_{\text{tx}} g_{i,b} \geq \Gamma_{\text{rsrp}}\}$ 
22:     if  $\mathcal{C}_i = \emptyset$  then
23:        $\mathcal{C}_i \leftarrow \mathcal{A}$ 
24:     end if
25:      $b_i \leftarrow \arg \max_{b \in \mathcal{C}_i} P_{i,b}$ 
26:      $z_{i,b_i} \leftarrow 1$ 
27:   end for
28:    $\rho_b \leftarrow \sum_{i \in \mathcal{U}} z_{i,b} d_i / R_{i,b}(\mathcal{A})$ 
29:    $\mathcal{V} \leftarrow \{i : R_{i,b_i} < R_{\min} \text{ or } P_{\text{tx}} g_{i,b_i} < \Gamma_{\text{rsrp}} \text{ or } \rho_{b_i} > \rho_{\max}\}$ 
30:   if  $\mathcal{V} = \emptyset$  or  $|\mathcal{A}| = B$  then
31:     break
32:   end if
33:    $b^+ \leftarrow \arg \max_{b \in \mathcal{B} \setminus \mathcal{A}} \sum_{i \in \mathcal{V}} d_i g_{i,b}$ 
34:    $\mathcal{A} \leftarrow \mathcal{A} \cup \{b^+\}$ 
35: until  $\mathcal{V} = \emptyset$  or  $|\mathcal{A}| = B$ 
36:  $\hat{\mathbf{a}}_b \leftarrow \mathbf{1}\{b \in \mathcal{A}\}$ ,  $\forall b$ 
37:  $\hat{z}_{i,b} \leftarrow z_{i,b}$ ,  $\forall i, b$ 
38: return  $\hat{\mathbf{Z}}, \hat{\mathbf{a}}$ 

```

Proof: Let $\mathbf{H}^{(\ell)}$ and $\tilde{\mathbf{H}}^{(\ell)}$ be the layer- ℓ hidden matrices generated from $\mathfrak{M}$ and $T_{\Pi}(\mathfrak{M})$, respectively. Then

$$\begin{aligned} \tilde{\mathbf{H}}^{(\ell)} &= \phi \left(\Pi \mathbf{S}^{(\ell)} \Pi^T \Pi \mathbf{X} \mathbf{W}^{(\ell)} \right) \\ &= \phi \left(\Pi \mathbf{S}^{(\ell)} \mathbf{X} \mathbf{W}^{(\ell)} \right) = \Pi \phi \left(\mathbf{S}^{(\ell)} \mathbf{X} \mathbf{W}^{(\ell)} \right) \\ &= \Pi \mathbf{H}^{(\ell)}. \end{aligned} \quad (37)$$

Since concatenation and multiplication by scalar α_{ℓ} act on feature coordinates and not on UE indices,

$$\parallel_{\ell \in \mathcal{L}} \alpha_{\ell} \tilde{\mathbf{H}}^{(\ell)} = \Pi \left(\parallel_{\ell \in \mathcal{L}} \alpha_{\ell} \mathbf{H}^{(\ell)} \right). \quad (38)$$

This proves (36). $\blacksquare$

Theorem 2 (Non-injectivity of uniform layer aggregation). *Let*

$$C(\mathfrak{M}) = (\mathbf{X}, \bar{\mathbf{S}}), \quad \bar{\mathbf{S}} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \mathbf{S}^{(\ell)}, \quad (39)$$

be the uniform layer aggregation map. If $|\mathcal{L}| \geq 2$, then C is not injective. Hence any predictor $H(C(\mathfrak{M}))$ cannot distinguish all multilayer inputs.

Proof: It is sufficient to construct two distinct inputs with the same image under C . Consider two layers p and q , and let $\mathbf{A} \neq \mathbf{B}$. Define

$$\begin{aligned} \mathfrak{M}_1 : \quad & \mathbf{S}^{(p)} = \mathbf{A}, \quad \mathbf{S}^{(q)} = \mathbf{B}, \\ \mathfrak{M}_2 : \quad & \mathbf{S}^{(p)} = \mathbf{B}, \quad \mathbf{S}^{(q)} = \mathbf{A}, \end{aligned} \quad (40)$$

with all remaining layers and $\mathbf{X}$ identical. Then

$$C(\mathfrak{M}_1) = C(\mathfrak{M}_2), \quad \mathfrak{M}_1 \neq \mathfrak{M}_2. \quad (41)$$

Thus C is non-injective. Consequently, every predictor that factors through C assigns the same output to $\mathfrak{M}_1$ and $\mathfrak{M}_2$. In contrast, the layer-aware encoder contains the ordered terms $\phi(\mathbf{S}^{(p)} \mathbf{X} \mathbf{W}^{(p)})$ and $\phi(\mathbf{S}^{(q)} \mathbf{X} \mathbf{W}^{(q)})$; for $\mathbf{W}^{(p)} \neq \mathbf{W}^{(q)}$, the two constructed inputs need not have identical embeddings. $\blacksquare$

Lemma 1 (Attention normalization). *Let α be defined by (29). Then*

$$\alpha \in \Delta^{|\mathcal{L}|-1} = \{\mathbf{u} \in \mathbb{R}_+^{|\mathcal{L}|} : \sum_{\ell \in \mathcal{L}} u_\ell = 1\}. \quad (42)$$

Proof: For all ℓ , $\exp(\eta_\ell) > 0$; therefore $\alpha_\ell \geq 0$. Moreover,

$$\sum_{\ell \in \mathcal{L}} \alpha_\ell = \frac{\sum_{\ell \in \mathcal{L}} \exp(\eta_\ell)}{\sum_{r \in \mathcal{L}} \exp(\eta_r)} = 1. \quad (43)$$

The layer-attention concentration reported in the experiments is measured by

$$H_\alpha = - \sum_{\ell \in \mathcal{L}} \alpha_\ell \log \alpha_\ell. \quad (44)$$

Theorem 3 (Finite monotone repair). *Let $\mathcal{A}^{(0)}$ be the active set initialized by (33). Algorithm 1 generates a sequence $\{\mathcal{A}^{(m)}\}_{m \geq 0}$ satisfying*

$$\mathcal{A}^{(0)} \subseteq \mathcal{A}^{(1)} \subseteq \dots \subseteq \mathcal{B}. \quad (45)$$

Moreover, the number of strict expansions is at most $B - |\mathcal{A}^{(0)}|$, and if the algorithm returns at iteration m with $\Phi(\hat{\mathbf{Z}}, \hat{\mathbf{a}}) = 1$, then $(\hat{\mathbf{Z}}, \hat{\mathbf{a}})$ is feasible for (18).

Proof: At any infeasible iteration, the update has the form

$$\mathcal{A}^{(m+1)} = \mathcal{A}^{(m)} \cup \{b_m^+\}, \quad b_m^+ \in \mathcal{B} \setminus \mathcal{A}^{(m)}. \quad (46)$$

Therefore (45) holds and each strict expansion increases $|\mathcal{A}^{(m)}|$ by one. Since $|\mathcal{B}| = B$, at most $B - |\mathcal{A}^{(0)}|$ strict expansions are possible. If the algorithm exits with $\Phi(\hat{\mathbf{Z}}, \hat{\mathbf{a}}) = 1$, then the association, activation, RSRP, rate, and load constraints hold by the definition of Φ . $\blacksquare$

TABLE II
BASELINE SIMULATION AND TRAINING CONFIGURATION

Parameter	Value
Number of BSs, B	7
Number of UEs, N	50
Service-region radius	650 m
Carrier frequency	3.5 GHz
Bandwidth	20 MHz
BS transmit power	20 W
Noise power	−94 dBm
Path-loss exponent	3.2
Shadowing standard deviation	6 dB
Minimum RSRP threshold	−115 dBm
Minimum rate, $R_{\min}$	1 Mbps
Traffic demand range	0.5–6 Mbps
Load limit, $\rho_{\max}$	1.0
BS fixed/dynamic/sleep power	90 W / 55 W / 8 W
BS switching cost, κ	35 W-equivalent per state change
Traffic periods, T	24
Training/test snapshots per seed	60 / 20
Random seeds	11, 13, 17, 19, 23, 29, 31, 37, 41, 43
Hidden dimension	24
Training epochs / patience	80 / 20
Learning rate	0.018

Corollary 1 (Oracle bound). *Let $\mathcal{C}$ be the finite candidate set searched by the oracle and let $J(\mathbf{Z}, \mathbf{a}) = P_{\text{net}}(\mathbf{Z}, \mathbf{a})$. If*

$$(\mathbf{Z}^*, \mathbf{a}^*) \in \arg \min_{(\mathbf{Z}, \mathbf{a}) \in \mathcal{C}, \Phi(\mathbf{Z}, \mathbf{a})=1} J(\mathbf{Z}, \mathbf{a}), \quad (47)$$

then for every feasible $(\mathbf{Z}, \mathbf{a}) \in \mathcal{C}$,

$$J(\mathbf{Z}^*, \mathbf{a}^*) \leq J(\mathbf{Z}, \mathbf{a}). \quad (48)$$

Proof: The inequality is the defining property of the minimum over the feasible subset of $\mathcal{C}$. $\blacksquare$

V. EXPERIMENTAL DESIGN

A. Simulation Scenario

The experimental study uses simulated dense downlink cellular snapshots so that the true channel, traffic, association, load, and energy quantities are available for controlled comparison. Each snapshot contains B BSs deployed in a circular service region and N UEs drawn uniformly over the same region. One BS is placed at the center and the remaining BSs are placed on a ring, producing a dense small-cell layout with overlapping coverage. UE traffic demands vary across space and time: each snapshot is assigned a traffic period $t \in \{0, \dots, T-1\}$, and the demand distribution combines lognormal user variation, a periodic traffic factor, and a spatial hotspot.

The downlink channel gain $g_{i,b}$ includes distance-dependent path loss, lognormal shadowing, and small-scale fading. Unless otherwise stated, all BSs transmit with the same power and share the same bandwidth. The all-on network is used only as a reference operating point; learned and heuristic policies are allowed to switch off BSs through the active-BS vector $\mathbf{a}$ defined in Section III. Fig. 3 illustrates the simulated deployment and the information flow from snapshot inputs to radio-channel generation, multilayer graph construction, and control evaluation. Table II lists the baseline simulation configuration used for the initial comparative study.

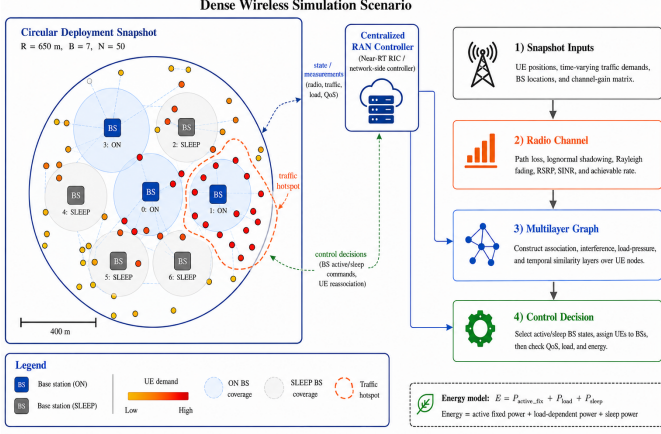


Fig. 3. Illustrative dense wireless control scenario considered in this study. A network-side controller collects radio, traffic, and load information from a dense BS cluster, constructs the multilayer UE graph, and jointly determines UE association and BS active/sleep states. When a BS is switched to sleep mode, its UEs are reassigned to feasible neighboring active BSs subject to QoS and load constraints. The scenario is motivated by HetNet BS switch-off and load-transfer models in [1], [2] and by network-side energy-saving control concepts in O-RAN [5].

B. Reference Decisions

Each training snapshot is labeled by a finite-search reference policy. The reference policy enumerates candidate active-BS subsets $\mathcal{A} \subseteq \mathcal{B}$ and assigns UEs greedily within each subset using channel quality, demand, and projected load. For each candidate, it computes an energy-penalized score

$$Q(\mathbf{Z}, \mathbf{a}) = P_{\text{net}}(\mathbf{Z}, \mathbf{a}) + \lambda_s (1 - \text{SR}(\mathbf{Z}, \mathbf{a})) + \lambda_p \left[\max_b \rho_b(\mathbf{Z}, \mathbf{a}) - \rho_{\max} \right]_+, \quad (49)$$

where $[x]_+ = \max(x, 0)$ and SR is the served ratio. The reference decision is the lowest-score feasible candidate in the searched set. This reference is not used as a claimed globally optimal solution of (18); it is used as a finite benchmark for supervised training and for measuring the energy gap of learned policies.

C. Compared Policies

The comparative study evaluates four groups of policies.

First, conventional association heuristics are included as non-learning baselines. The RSRP policy assigns each UE to the BS with maximum received power. The SINR policy assigns each UE to the BS with maximum all-on SINR. The load-aware policy sequentially assigns high-demand UEs to BSs that minimize projected load while maintaining radio feasibility when possible. These heuristics are repaired before evaluation using the same QoS/load guard applied to learned policies, so the comparison measures the quality of the association preference rather than failure of a postprocessing step.

Second, explicit sleep-control baselines are included to avoid comparing against association-only heuristics. RSRP-sleep, SINR-sleep, and load-sleep start from the corresponding association rule and greedily deactivate BSs when the resulting

repaired association remains feasible and reduces energy. Greedy-sleep is a more demanding baseline that uses the same energy-aware assignment rule as the finite-search reference but performs backward one-cell pruning instead of enumerating all active-BS subsets. These baselines evaluate whether a learned policy improves over hand-designed BS switch-off logic.

Third, single-representation learning baselines are trained on the same reference labels. The flat MLP receives UE features only and does not use graph propagation. The aggregated GCN uses the average support

$$\bar{\mathbf{S}} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \mathbf{S}^{(\ell)} \quad (50)$$

and therefore receives a single graph representation. These two baselines separate the value of graph propagation from the value of preserving separate relation layers.

Fourth, the proposed multilayer models are evaluated. The fixed-fusion ML-GCN uses (26)–(27), while the attention ML-GCN uses (29)–(30). For each learned representation, we report the fixed top- K policy, the score-guided pruned policy, and the stability-aware pruned policy. A finite-search reference policy is reported as an upper-bound benchmark within the searched candidate class.

D. Training Protocol

All learning models are trained by minimizing the association loss in (31) over the training snapshots. Node features are standardized using the training set statistics and the same transformation is applied to the test snapshots. The active-BS selector is trained after the association model by using the model probability outputs and the reference active sets. The number of selected active BSs is initialized from the rounded mean active-BS count of the reference decisions in the training set.

For each test snapshot, a learned model first outputs UE-to-BS probabilities, then active-BS scores, then an initial association over the selected active set. Algorithm 1 finally checks RSRP, rate, and load constraints and activates additional BSs only when the selected set cannot support the snapshot. All reported metrics are computed after this final repaired decision.

E. Evaluation Metrics

The primary metric is network energy. Let P_{all} denote the energy consumed when all BSs are active under the same power model. The energy saving of a policy π is

$$\text{ES}_\pi = \frac{P_{\text{all}} - P_\pi}{P_{\text{all}}}, \quad (51)$$

where $P_\pi = P_{\text{net}}(\mathbf{Z}_\pi, \mathbf{a}_\pi)$. The energy gap relative to the reference policy is

$$\text{Gap}_\pi = \frac{P_\pi - P_{\text{ref}}}{P_{\text{ref}}}, \quad (52)$$

where P_{ref} is the reference energy for the same snapshot.

Constraint satisfaction is measured by served ratio,

$$\text{SR}_\pi = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left\{ P_{\text{tx}} g_{i,b_i} \geq \Gamma_{\text{rsrp}}, \right. \\ \left. R_{i,b_i} \geq R_{\min}, \rho_{b_i} \leq \rho_{\max} \right\}. \quad (53)$$

Operational compactness and load behavior are reported using active-BS count

$$A_\pi = \sum_{b \in \mathcal{B}} a_{\pi,b}, \quad (54)$$

maximum load

$$\rho_{\max}^\pi = \max_{b \in \mathcal{B}} \rho_b, \quad (55)$$

and mean active load

$$\bar{\rho}_{\text{on}}^\pi = \frac{\sum_{b \in \mathcal{B}} a_{\pi,b} \rho_b}{\sum_{b \in \mathcal{B}} a_{\pi,b}}. \quad (56)$$

Assignment accuracy against reference labels is reported only as a diagnostic, because a policy can have low label agreement while still producing a feasible low-energy association.

F. Layer Ablation and Stress Tests

To evaluate the role of each relation layer, the experiments include single-layer and leave-one-layer-out variants of the ML-GCN. For a layer ℓ , the leave-one-layer-out energy effect is

$$\Delta_\ell^{(-)} = \text{ES}_{\text{full}} - \text{ES}_{-\ell}, \quad (57)$$

and the single-layer contribution is

$$\Delta_\ell^{(+)} = \text{ES}_\ell - \text{ES}_{\text{flat}}. \quad (58)$$

Positive $\Delta_\ell^{(-)}$ indicates that removing layer ℓ reduces energy saving, while positive $\Delta_\ell^{(+)}$ indicates that layer ℓ alone improves over a feature-only model.

Pairwise layer redundancy is measured before training using support-matrix dissimilarity,

$$D_{\ell,r} = \frac{\|\mathbf{S}^{(\ell)} - \mathbf{S}^{(r)}\|_F}{\|\mathbf{S}^{(\ell)}\|_F + \epsilon}, \quad \ell \neq r, \quad (59)$$

where $\epsilon > 0$ avoids division by zero. Attention concentration is measured using (44). In addition to the baseline setting in II, stress tests vary bandwidth, QoS threshold, UE density, and shadowing variance.

VI. RESULTS AND DISCUSSION

A. Static Energy Versus Switching-Aware Control

Table III reports the main ten-seed simulation comparison. The focus is the operational comparison between greedy sleep control and the proposed stability-aware multilayer graph controllers. Greedy sleep is a non-learning BS switch-off baseline that directly removes cells when one-cell pruning reduces snapshot energy. However, the simulation results show that the proposed stability-aware ML-GCN improves over greedy sleep under both static and switching-adjusted energy while preserving full service feasibility.

The stability-aware ML-GCN search policy achieves 79.0% static energy saving and 76.1% switching-adjusted energy

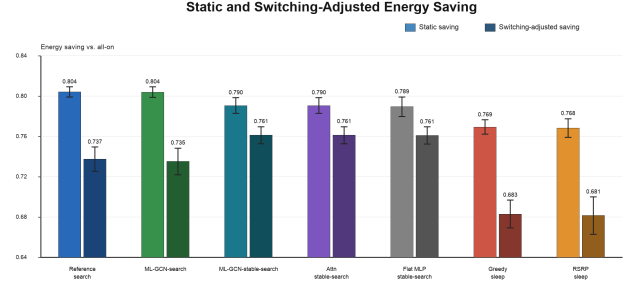


Fig. 4. Static and switching-adjusted energy saving across the main policies. Stability-aware ML-GCN search has higher saving than greedy sleep under both reported energy metrics while preserving full served ratio.

saving relative to the all-on network. Greedy sleep achieves 76.9% static saving and 68.3% switching-adjusted saving. Thus, the learned multilayer controller reduces static energy while also reducing the switching penalty caused by active-BS churn. The improvement is not obtained by relaxing constraints: both stability-aware ML-GCN search and greedy sleep maintain served ratio 1.0. The main operational difference is that stability-aware ML-GCN search uses fewer BS state transitions, reducing the mean number of state changes from 2.50 for greedy sleep to 0.85 per snapshot.

The finite-search reference is retained as a bounded reference and training signal. It gives the largest static saving among the searched candidates, whereas stability-aware ML-GCN search gives the largest switching-adjusted saving among the implementable policies in this simulation comparison. This distinction supports the modeling hypothesis: preserving multilayer wireless structure and applying stability-aware active-cell control can improve energy-efficient BS switch-off when both static energy and switching behavior are reported.

B. Effect of Stability-Aware Pruning

Fig. 5 reports the BS state-change behavior. Stability-aware pruning reduces active-cell churn by reusing the previous active set when it remains feasible and accepting a state change only when the switching-adjusted objective improves or feasibility requires it. In the simulation results, ML-GCN-stable-search has fewer mean state changes than both ML-GCN-pruned and greedy sleep.

Relative to score-guided ML-GCN-pruned, the stable-search version increases switching-adjusted saving from 73.4% to 76.1% while reducing state changes from 1.54 to 0.85 per snapshot. Relative to greedy sleep, ML-GCN-stable-search reduces mean state changes by approximately 1.65 per snapshot. The reduction is operationally relevant because BS ON/OFF transitions can incur signaling, hardware, and service-continuity costs. The result also identifies the role of the decision layer: multilayer graph learning provides association preferences from heterogeneous wireless relations, while stability-aware pruning converts those preferences into compact and temporally smoother active-BS sequences.

TABLE III
TEN-SEED PERFORMANCE WITH SWITCHING COST

Policy	Switching saving	Static saving	Switches	Served ratio	Active BSs
Reference search	0.7372 ± 0.0121	0.8040 ± 0.0050	1.938 ± 0.225	1.000	1.279
ML-GCN-search	0.7350 ± 0.0133	0.8038 ± 0.0052	1.996 ± 0.253	1.000	1.279
ML-GCN-stable-search	0.7610 ± 0.0084	0.7905 ± 0.0076	0.854 ± 0.177	1.000	1.400
Attn-ML-GCN-stable-search	0.7610 ± 0.0085	0.7904 ± 0.0077	0.854 ± 0.177	1.000	1.400
Flat MLP-stable-search	0.7607 ± 0.0086	0.7893 ± 0.0097	0.829 ± 0.197	1.000	1.408
Greedy sleep	0.6828 ± 0.0138	0.7692 ± 0.0071	2.504 ± 0.308	1.000	1.721
ML-GCN-pruned	0.7338 ± 0.0123	0.7868 ± 0.0073	1.538 ± 0.256	1.000	1.458
Attn-ML-GCN-pruned	0.7338 ± 0.0124	0.7869 ± 0.0073	1.538 ± 0.256	1.000	1.458
RSRP sleep	0.6813 ± 0.0184	0.7682 ± 0.0092	2.521 ± 0.279	1.000	1.742
RSRP	0.3223 ± 0.0034	0.3230 ± 0.0035	0.021 ± 0.035	1.000	6.988

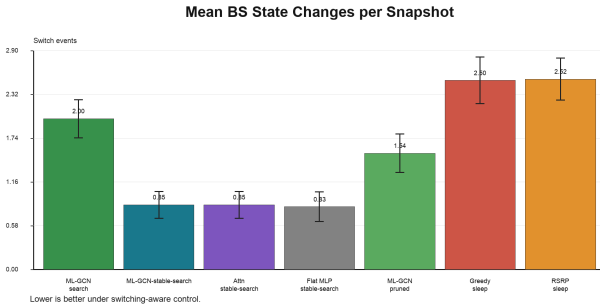


Fig. 5. Mean BS state changes per snapshot. Stability-aware ML-GCN reduces active-cell churn compared with greedy sleep and score-guided pruning.

C. Switching-Cost Sensitivity

Fig. 6 shows the sensitivity to the BS switching penalty κ . When switching is free, policies with aggressive pruning can perform well because the objective reduces to static snapshot energy. As κ increases, the ranking shifts toward policies that avoid unnecessary active-cell transitions. Around the operating value $\kappa = 35$ W-equivalent used in the baseline configuration, ML-GCN-stable-search is above greedy sleep and the other learned baselines under switching-adjusted saving.

This sensitivity analysis evaluates the effect of transition cost on BS switch-off decisions. A controller that saves static energy by rapidly changing BS states can lose part of its benefit when switching overhead is included. The proposed stable-search ML-GCN preserves most of the static energy benefit while reducing state-change cost, yielding the largest switching-adjusted saving in the simulation comparison.

D. Multilayer Representation and Learned Baselines

The results are reported for the complete multilayer graph-control pipeline rather than for assignment accuracy alone. The flat MLP uses UE features without graph propagation, while the aggregated GCN collapses heterogeneous wireless relations into one support. The multilayer GCN preserves association competition, interference similarity, load pressure, and temporal traffic similarity as separate relation channels before active-cell selection and repair.

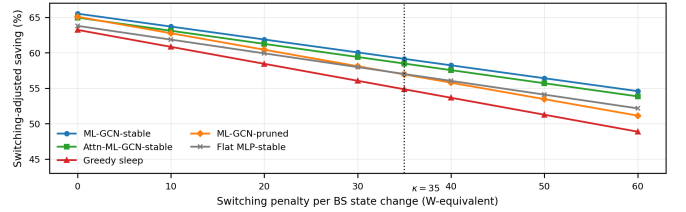


Fig. 6. Sensitivity to the BS switching penalty. Stable multilayer graph control provides higher switching-adjusted saving as active-cell transitions become costly.

Table IV separates the effect of representation from the effect of the active-set decision layer. ML-GCN-search reaches the largest static saving among learned online policies because it closely follows the finite-search reference, but it changes BS states more frequently. Adding stability reduces state changes and improves switching-adjusted saving. The comparison with Flat MLP-stable-search quantifies the contribution of the stable active-set controller to switching-aware performance. The multilayer representation contributes by preserving relation-aware association preferences and by maintaining comparable performance across the static, switching, ablation, dissimilarity, and robustness views, rather than by acting as an isolated classifier.

The ML-GCN-stable-search policy obtains the largest switching-adjusted saving among the multilayer policies in the simulation comparison because it combines three ingredients: relation-aware association preferences, compact active-BS search, and switching-aware stability. The pruned ML-GCN has comparable static saving, but it changes BS states more often than the stable version. The attention ML-GCN also has comparable performance but remains slightly below the fixed-fusion ML-GCN in this run, suggesting that the present global attention mechanism may still be conservative. Snapshot-dependent attention remains a potential extension.

E. Runtime and Complexity

Table V reports CPU timing and online decision complexity for the revised policies. The benchmark uses $N = 50$ UEs, $B = 7$ BSs, 12 training snapshots, and 5 test snapshots for timing. Let L denote the number of relation layers, d the input feature dimension, h the hidden dimension, and R the

TABLE IV
COMPONENT ABLATION: REPRESENTATION VERSUS ACTIVE-SET CONTROL

Policy	Main component tested	Static saving	Switching saving	Switches
Greedy sleep	Hand-designed active-cell pruning	0.7692 ± 0.0071	0.6828 ± 0.0138	2.504 ± 0.308
ML-GCN-search	Multilayer representation with search, no stability inertia	0.8038 ± 0.0052	0.7350 ± 0.0133	1.996 ± 0.253
ML-GCN-stable	Multilayer representation with stability-aware pruning	0.7857 ± 0.0074	0.7469 ± 0.0098	1.125 ± 0.196
Flat MLP-stable-search	Feature-only representation with stable search	0.7893 ± 0.0097	0.7607 ± 0.0086	0.829 ± 0.197
ML-GCN-stable-search	Multilayer representation with stable search	0.7905 ± 0.0076	0.7610 ± 0.0084	0.854 ± 0.177

cost of one repair/evaluation pass. Offline training is reported separately because it is amortized before deployment.

The runtime results support the switching-aware comparison. Greedy sleep is a hand-designed controller, but its online latency is higher than ML-GCN-stable and attention-ML-GCN-stable in the timing benchmark. The finite-search reference is slower and scales exponentially with B ; it is therefore used as a bounded reference and training signal rather than as an online controller. The stable learned policies therefore provide an intermediate operating point: they have fewer BS state changes than greedy sleep under switching cost and lower inference latency than greedy sleep and finite search in the timing benchmark.

F. Representative Snapshot Behavior

Fig. 7 shows one representative test snapshot. The left panel gives the all-on SINR matrix, sorted by the reference association. The center and right panels compare the reference association and the ML-GCN association on the same UE/BS geometry. Both decisions preserve full service, but the learned policy uses one additional active BS in this snapshot. This is consistent with the aggregate results: the learned controller approaches the reference switch-off behavior but remains more conservative in order to satisfy QoS and load constraints after repair.

The snapshot also illustrates why assignment accuracy is not sufficient as a primary metric. A UE can be moved from one feasible BS to another and still produce a valid low-energy decision. Therefore, the correct evaluation target is not exact label reproduction; it is constrained network control measured by energy, active BS count, load, and served ratio.

G. Bandwidth and QoS Sensitivity

Fig. 8 evaluates bandwidth and QoS sensitivity. At 10 MHz, the network is highly resource constrained; all methods show low served ratio and energy saving is limited because most BSs must remain active. As bandwidth increases, the reference and learned policies exploit the additional capacity by switching off more BSs. At 20 MHz, ML-GCN reaches 51.5% energy saving with full served ratio, while RSRP and load-aware association remain near 17% because they do not actively compact the BS set. At 80 MHz, the learned controllers exceed 80% energy saving and approach the reference trend.

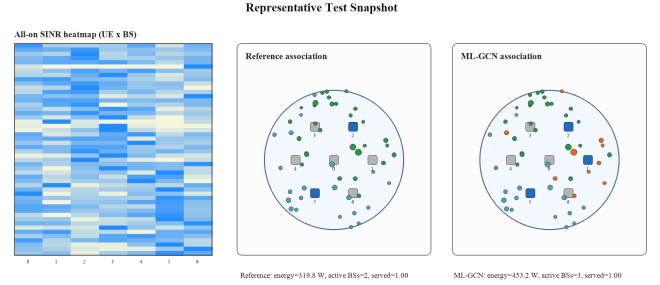


Fig. 7. Representative test snapshot. The heatmap shows all-on SINR over UE-BS pairs; the association maps show the finite-search reference and the ML-GCN decision after QoS/load repair. The learned policy maintains full service while selecting a compact active-BS set.

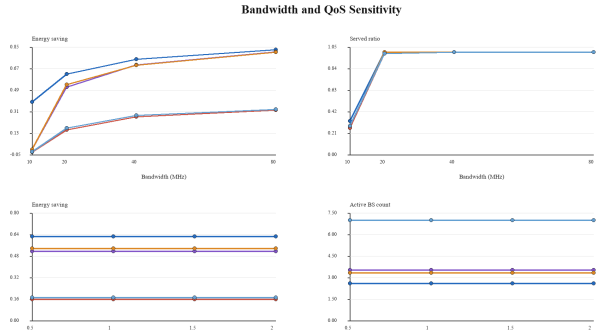


Fig. 8. Bandwidth and QoS sensitivity. Increasing bandwidth creates additional opportunity for BS switch-off, while the tested QoS range is not the dominant limiting factor under the baseline traffic load.

The QoS sweep is comparatively flat under the tested range $R_{\min} \in [0.5, 2]$ Mbps. This indicates that, in the baseline traffic and bandwidth setting, the dominant limiting factor is not the minimum-rate threshold itself but the load and active-cell structure created by the association policy. More aggressive QoS stress would require higher demand or lower bandwidth if rate feasibility is to become the dominant bottleneck.

To complement the one-dimensional sweeps, Fig. 9 reports a joint QoS-density stress surface for ML-GCN under three shadowing levels. The surface confirms that UE density is the dominant stressor in the tested regime: energy saving remains high when the network has 30–50 UEs, but decreases sharply

TABLE V
RUNTIME AND ONLINE COMPLEXITY BENCHMARK

Policy	Offline fit (s)	Median latency (ms)	Online decision complexity
RSRP	—	3.48	$\mathcal{O}(NB)$
Load-aware	—	5.13	$\mathcal{O}(NB + N \log N)$
Load-sleep	—	36.16	$\mathcal{O}(B^2 N + BN \log N)$
Greedy sleep	—	84.17	$\mathcal{O}(B^2 N + BN \log N)$
ML-GCN	0.21	4.98	$\mathcal{O}(\sum_{\ell} E_{\ell} d + LNd h + LN h B)$
ML-GCN-pruned	0.21	40.06	$\mathcal{O}(\sum_{\ell} E_{\ell} d + LNd h + LN h B + BR)$
ML-GCN-stable	0.21	51.50	$\mathcal{O}(\sum_{\ell} E_{\ell} d + LNd h + LN h B + BR)$
Attn-ML-GCN-stable	0.31	41.20	$\mathcal{O}(\sum_{\ell} E_{\ell} d + LNd h + LN h B + BR)$
Reference search	—	228.75	$\mathcal{O}(2^B NB)$

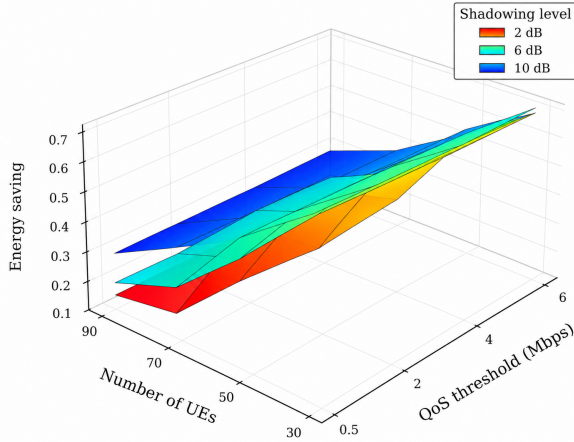


Fig. 9. Joint QoS-density surface for the proposed ML-GCN. Each surface reports mean energy saving after QoS/load repair for a different shadowing standard deviation. UE density has a larger effect on energy saving than the tested QoS-threshold variation.

as the UE count grows to 70–90. Changes in the QoS threshold produce a smaller effect than density, which is consistent with the earlier QoS sweep. This behavior indicates that the learned controller is mainly limited by load concentration and active-cell capacity under dense traffic, rather than by the minimum-rate threshold alone.

H. Scalability and Channel Robustness

Fig. 10 studies UE-density and shadowing sweeps. As the number of UEs increases from 30 to 90, all policies experience reduced energy saving because additional traffic requires more active BSs. The reference decreases from 73.8% to 39.2% energy saving, while ML-GCN decreases from 70.5% to 20.6%. The gap widens in the dense-load regime, which suggests that the current learned active-BS selector is conservative when many UEs compete for limited resources. Nevertheless, ML-GCN remains above RSRP and load-aware association across the density sweep.

The shadowing sweep shows that the learned policies are stable under channel uncertainty in the tested range. ML-GCN maintains full served ratio and approximately 51–55% energy saving as the shadowing standard deviation varies from 2 to 14 dB. The reference energy saving also remains close to 62–63%. This robustness is expected because the multilayer

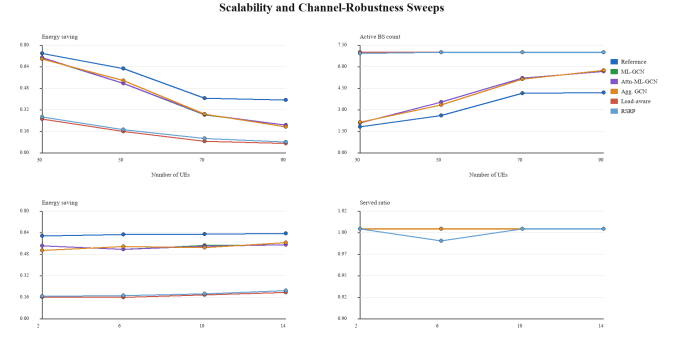


Fig. 10. Scalability and channel-robustness sweeps. Higher UE density reduces energy-saving opportunity and increases active BS count. Under the tested shadowing levels, the learned policies preserve service feasibility and remain more energy-efficient than conventional association rules.

representation uses multiple relational views: channel-profile similarity, association competition, demand/load structure, and temporal similarity. If one view becomes noisier, the remaining layers still provide discriminative structure for active-cell selection and association repair.

I. Layer Ablation

Fig. 11 reports layer ablation. Removing the load layer has the largest leave-one-layer-out penalty: the energy saving decreases from 55.4% to 53.2%, and the active-BS count increases from 3.2 to 3.4. This confirms that demand/load structure is relevant to deciding which BSs can remain active without overload. Removing the association, interference, or temporal layer also reduces energy saving, although with smaller penalties.

The single-layer comparison gives a complementary view. The interference layer alone gives the largest single-layer improvement over the feature-only baseline, while association, load, and temporal layers have smaller individual effects but contribute jointly. Taken together, these two ablation views support the modeling hypothesis that no single layer fully explains the energy-aware association decision. The multilayer model benefits from preserving heterogeneous dependencies before fusion.

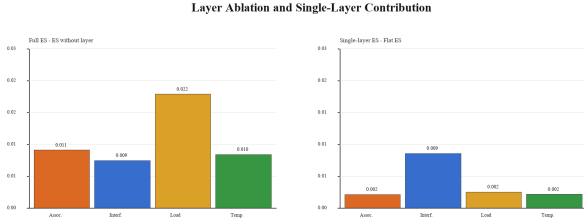


Fig. 11. Layer ablation analysis. The load layer has the largest leave-one-layer-out effect, while the interference layer gives the largest single-layer gain over the feature-only baseline.

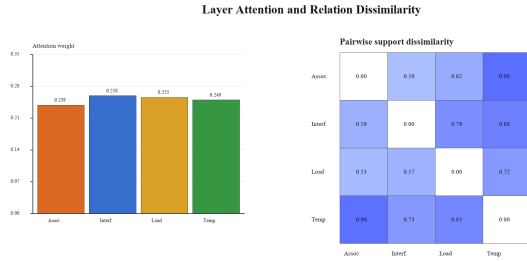


Fig. 12. Layer attention and relation dissimilarity. Attention weights are nearly balanced across relation layers, while the support matrices remain structurally distinct.

J. Attention and Relation Dissimilarity

Fig. 12 shows the learned attention distribution and pairwise support-matrix dissimilarity. The attention model assigns comparable weights to the four layers, with a mild preference for interference and load. This explains why attention and fixed fusion perform similarly in the main comparison: the current global attention mechanism does not strongly suppress any layer. A more expressive snapshot-dependent attention mechanism may differentiate layers more sharply under heterogeneous traffic regimes.

The dissimilarity heatmap shows that the temporal layer is the most distinct from the association layer, while association and interference have moderate overlap. The nonzero off-diagonal dissimilarities support the use of separate relation layers rather than a single aggregated support. Even when attention weights are nearly uniform, the layer-specific filters still process different support matrices before fusion.

Layer dissimilarity and layer ablation indicate that the association, interference, load, and temporal supports are not interchangeable projections of the same graph. The control behavior therefore depends on which network relations are preserved before learning. In the BS switch-off setting, the controller acts on infrastructure states, while feasibility and stability are determined by heterogeneous UE-UE relations.

K. Discussion

The experiments lead to four observations. First, association-only heuristics do not compact the active-BS set and therefore provide limited energy savings relative

to sleep-control policies. Second, greedy sleep is an active-cell pruning baseline for evaluating learned BS switch-off methods. Third, switching cost changes the operational ranking because policies with frequent active-cell transitions lose part of their static-energy benefit. Fourth, the proposed method is formulated as a multilayer graph-control framework with feasibility repair and active-cell stability, rather than as an association-label classifier.

The results indicate that BS switch-off evaluation should include static energy, switching-adjusted energy, served ratio, load feasibility, active-BS count, and online decision latency. The proposed framework addresses these criteria through the combination of multilayer representation learning, active-cell selection, stability-aware pruning, and QoS/load repair. The component ablation also indicates that the decision layer contributes measurably to switching-aware performance, while the multilayer representation provides a relation-aware and permutation-equivariant model of the wireless dependencies.

VII. CONCLUSION

This paper developed a switching-aware multilayer graph-learning framework for energy-efficient user association and BS switch-off in dense wireless networks. Each wireless snapshot was represented as a UE-level multilayer graph with separate association, interference, load, and temporal relation layers. These layers were processed by fixed-fusion and attention-weighted ML-GCN models, and the learned association preferences were converted into feasible network decisions by active-cell selection, score-guided pruning, stability-aware pruning, and QoS/load repair.

The primary empirical finding is that the value of a BS switch-off policy depends on whether active-cell transitions are counted. Greedy sleep control is a relevant baseline, but in the simulation comparison the stability-aware ML-GCN improves over it under both energy criteria. Stability-aware ML-GCN search achieves 79.0% static energy saving and 76.1% switching-adjusted saving, compared with 76.9% and 68.3% for greedy sleep, while maintaining served ratio 1.0. The improvement is supported by lower active-cell churn: ML-GCN-stable-search reduces mean BS state changes from 2.50 to 0.85 per snapshot. These results support the proposed multilayer graph-control framework as an energy-efficient and switching-aware approach to dense-network BS sleep control. Runtime measurements showed that the stable learned policies have lower online latency than greedy sleep and lower latency than finite active-cell search. The component ablation indicates that, in the present implementation, the stability-aware decision layer provides the largest contribution to switching-adjusted performance, while the multilayer representation provides a relation-aware and permutation-equivariant model for heterogeneous wireless dependencies.

The study has several limitations. First, the evaluation is simulation based and uses controlled channel, traffic, and power-model assumptions. These assumptions support reproducible comparison, but they do not capture all operational details of deployed radio access networks. Second, the finite reference search is tractable for the considered BS scale and

is used as a bounded reference rather than as a scalable optimal solver. Third, the current attention mechanism uses global layer weights; it does not yet adapt relation importance on a snapshot-by-snapshot basis. Fourth, the stability-aware pruning rule is a post-learning decision layer, so switching cost is not yet optimized end-to-end inside the GNN training objective.

Future work will extend the framework in four directions. First, the simulation environment should be expanded to include irregular BS deployments, mobility traces, nonuniform traffic hotspots, and more detailed switching delays and signaling costs. Second, adaptive attention and layer-selection mechanisms should be developed so that relation importance can change with traffic load, channel uncertainty, and topology. Third, switching cost and feasibility constraints should be integrated more directly into training through constrained learning, differentiable repair, or safe reinforcement learning. Fourth, transfer across network sizes and operating regimes should be evaluated so that models trained on one deployment can generalize to different densities, bandwidths, and service requirements. These extensions would move the proposed switching-aware multilayer graph framework closer to practical energy-efficient control in future intelligent wireless networks.

DATA AND CODE AVAILABILITY

The source code, simulation configurations, generated result files, MATLAB figure scripts, and manuscript assets used in this study are publicly available at <https://github.com/kazim01011/multilayer-ua-energy>. The repository contains the simulator, multilayer graph construction routines, learning models, baseline policies, reproducibility scripts, and aggregate CSV outputs used to generate the reported tables and figures.

REFERENCES

- [1] H. Ghazzai, M. J. Farooq, A. Alsharoa, E. Yaacoub, A. Kadri, and M.-S. Alouini, "Green networking in cellular HetNets: A unified radio resource management framework with base station ON/OFF switching," *IEEE Transactions on Vehicular Technology*, vol. 66, no. 7, pp. 5879–5893, 2017.
- [2] E. Oh, K. Son, and B. Krishnamachari, "Dynamic base station switching-on/off strategies for green cellular networks," *IEEE Transactions on Wireless Communications*, vol. 12, no. 5, pp. 2126–2136, 2013.
- [3] M. Feng, S. Mao, and T. Jiang, "Base station ON-OFF switching in 5G wireless networks: Approaches and challenges," *IEEE Wireless Communications*, vol. 24, no. 4, pp. 46–54, 2017.
- [4] OECD, "The environmental sustainability of communication networks," OECD, Tech. Rep., 2025. [Online]. Available: https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/02/the-environmental-sustainability-of-communication-networks_66fe3a2a/d1cb2210-en.pdf
- [5] O-RAN Alliance, "Potential energy savings features in O-RAN," White Paper, O-RAN.WP.Potential Energy Savings Features-v01.00, 2025. [Online]. Available: <https://mediastorage.o-ran.org/white-papers/O-RAN.SuFG.Potential%20Energy%20Savings%20Features%20in%20O-RAN%20white%20paper%202025-01.pdf>
- [6] M. Eisen and A. Ribeiro, "Optimal wireless resource allocation with random edge graph neural networks," *IEEE Transactions on Signal Processing*, vol. 68, pp. 2977–2991, 2020.
- [7] Y. Shen, Y. Shi, J. Zhang, and K. B. Letaief, "Graph neural networks for scalable radio resource management: Architecture design and theoretical analysis," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 1, pp. 101–115, 2021.
- [8] M. Kivela, A. Arenas, M. Barthelemy, J. P. Gleeson, Y. Moreno, and M. A. Porter, "Multilayer networks," *Journal of Complex Networks*, vol. 2, no. 3, pp. 203–271, 2014.
- [9] M. De Domenico, A. Sol 'e-Ribalta, E. Cozzo, M. Kivela, Y. Moreno, M. A. Porter, S. Gomez, and A. Arenas, "Mathematical formulation of multilayer networks," *Physical Review X*, vol. 3, p. 041022, 2013.
- [10] S. Boccaletti, G. Bianconi, R. Criado, C. I. del Genio, J. Gomez-Gardenes, M. Romance, I. Sendina-Nadal, Z. Wang, and M. Zanin, "The structure and dynamics of multilayer networks," *Physics Reports*, vol. 544, no. 1, pp. 1–122, 2014.
- [11] M. De Domenico, V. Nicosia, A. Arenas, and V. Latora, "Structural reducibility of multilayer networks," *Nature Communications*, vol. 6, p. 6864, 2015.
- [12] M. De Domenico, A. Sol 'e-Ribalta, E. Omodei, S. Gomez, and A. Arenas, "Ranking in interconnected multilayer networks reveals versatile nodes," *Nature Communications*, vol. 6, p. 6868, 2015.
- [13] J. Mirzaei, J. Mitra, and G. Poitau, "Graph attention network for optimal user association in wireless networks," 2025.
- [14] L. Lv, P. Hu, D. Bardou, T. Zhang, and Z. Zheng, "Community detection in multilayer networks via semi-supervised joint symmetric nonnegative matrix factorization," *IEEE Transactions on Network Science and Engineering*, vol. 10, no. 3, pp. 1623–1635, 2023.
- [15] M. Teng, Z. Yin, J. Huang, C. Gao, X. Li, V. I. Nekorkin, and Z. Wang, "Contrastive learning for multi-layer network community detection via learnable network augmentation," *IEEE Transactions on Network Science and Engineering*, vol. 12, no. 5, pp. 4227–4238, 2025.
- [16] P. Basaras, G. Iosifidis, D. Katsaros, and L. Tassioulas, "Identifying influential spreaders in complex multilayer networks: A centrality perspective," *IEEE Transactions on Network Science and Engineering*, 2019.
- [17] Z. An, X. Hu, Z. Wang, R. Jiang, J. Jiang, D. Chen, and Y. Jiang, "Identifying key nodes in multi-layer networks through relationship strength gravity centrality based on the multi-feature fusion," *IEEE Transactions on Network Science and Engineering*, 2026.
- [18] Y. Zhang, P. Wang, J. Lu, and T. Zhou, "Quantifying network dissimilarity via augmented networks," *IEEE Transactions on Network Science and Engineering*, vol. 13, pp. 3566–3579, 2026.
- [19] M. Grassia, M. De Domenico, and G. Mangioni, "mGNN: Generalizing the graph neural networks to the multilayer case," 2021.
- [20] X.-J. Zhang, X. Ding, H.-F. Zhang, D.-H. Pan, and K. Zhong, "A flexible monitoring framework via dynamic-multilayer graph convolution network," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–11, 2023.
- [21] S. Zandi, K. Korangi, M. 'Oskarsd 'ottir, C. Mues, and C. Bravo, "Attention-based dynamic multilayer graph neural networks for loan default prediction," *European Journal of Operational Research*, vol. 321, no. 2, pp. 586–599, 2025.
- [22] P. Zhang, Y. Zhang, J. Wang, and B. Yin, "MVMA-GCN: Multi-view multi-layer attention graph convolutional networks," *Engineering Applications of Artificial Intelligence*, vol. 126, p. 106717, 2023.